Automated Re-Training Pipeline Triggered by Data Drift

Assignment 8 | Demand Forecasting | MLOps PBA2

Student Details

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GitHub:	github.com/tanmays0/ADT23SOCB1617-drift-retraining-pipeline

Problem Statement

Domain: Demand Forecasting
Tech Stack: Prefect or Airflow, Evidently AI, Scikit-Learn, MLflow

Build an automated pipeline that continuously monitors incoming data streams for data and concept drift using Evidently AI. When drift exceeds a defined threshold (e.g., Wasserstein distance > 0.1), automatically trigger a re-training DAG.

Key Tasks:
1. Generate synthetic drifted data.
2. Register newly trained models into an MLflow Registry.
3. Set up alert notifications (Email).

Tech Stack (Implemented)

Prefect | Evidently AI | Scikit-Learn | MLflow | Email alerts (local files)

Key Results

- Drift detected: True (4/6 columns; share 0.667)
- Wasserstein (units_sold): 0.817 | price: 3.04 | temperature: 1.54
- Model registered: demand-forecast-model v1 (trigger=data_drift)
- Metrics: MAE ~ 3.52, R2 ~ 0.95
- Alert: email notification written to outputs/alerts/

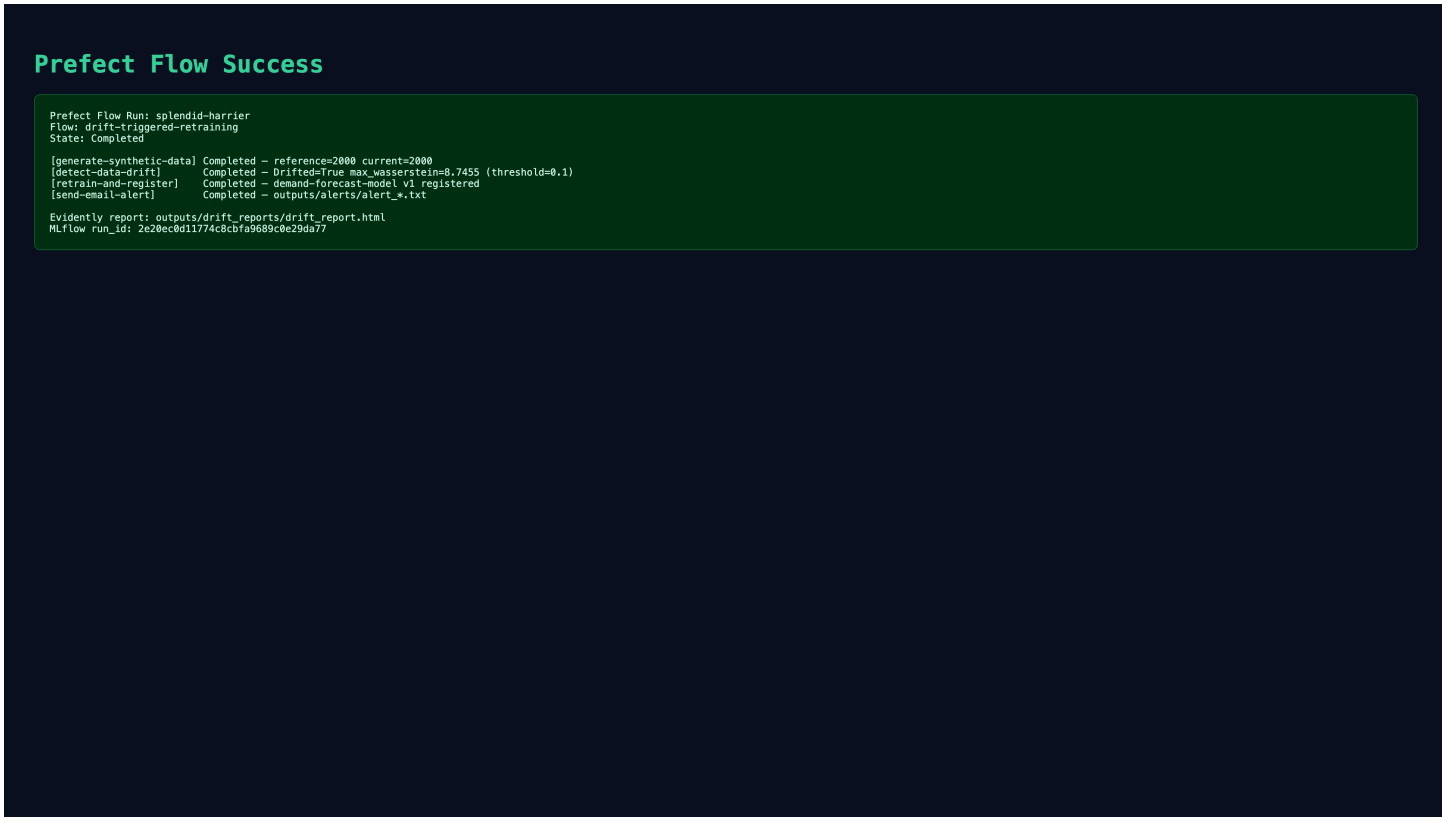
Where to check live output

GitHub: https://github.com/tanmays0/ADT23SOCB1617-drift-retraining-pipeline
MLflow UI: http://127.0.0.1:5000/#/models
Evidently report: outputs/drift_reports/drift_report.html
Email alert: outputs/alerts/alert_*.txt
Re-run: python scripts/run_demo.py

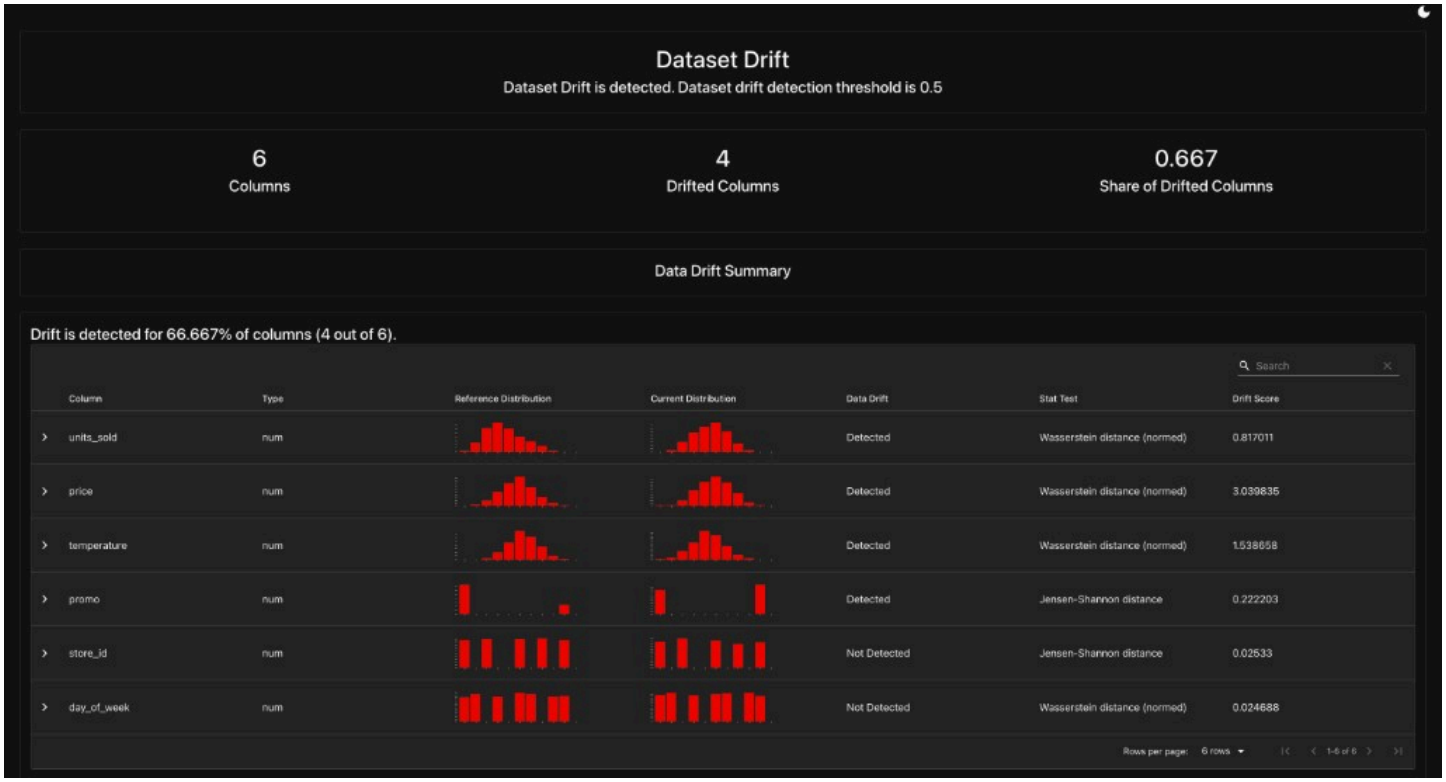
Screenshot Index

1. Prefect Flow Success
2. Report Screenshots (Evidently AI) - Dataset Drift Summary, column drift, values, correlations, distribution
3. MLflow Model Registry / Run (demand-forecast-model v1)
4. Email Alert Notification
5. GitHub Repository

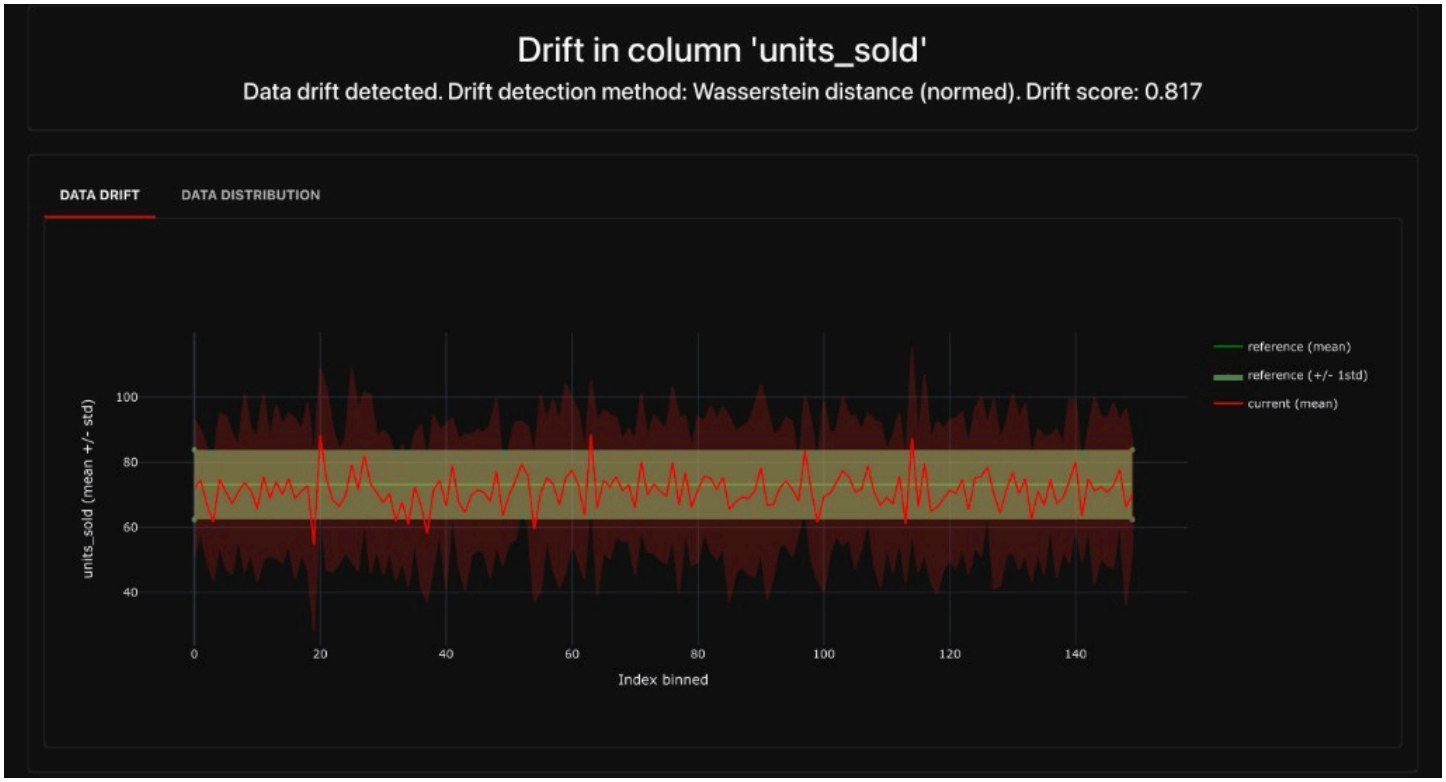
1. Prefect Flow Success - drift-triggered retrain completed



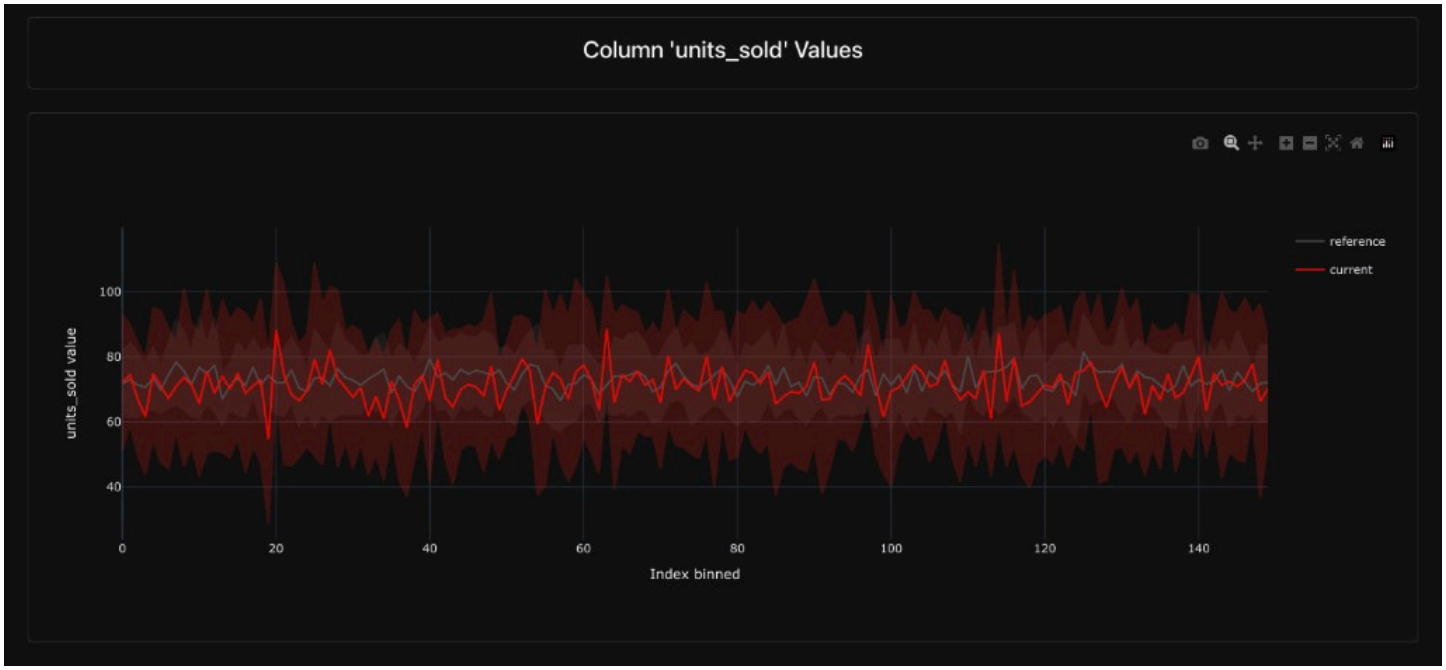
2. Report Screenshots - Dataset Drift Summary (Evidently AI)



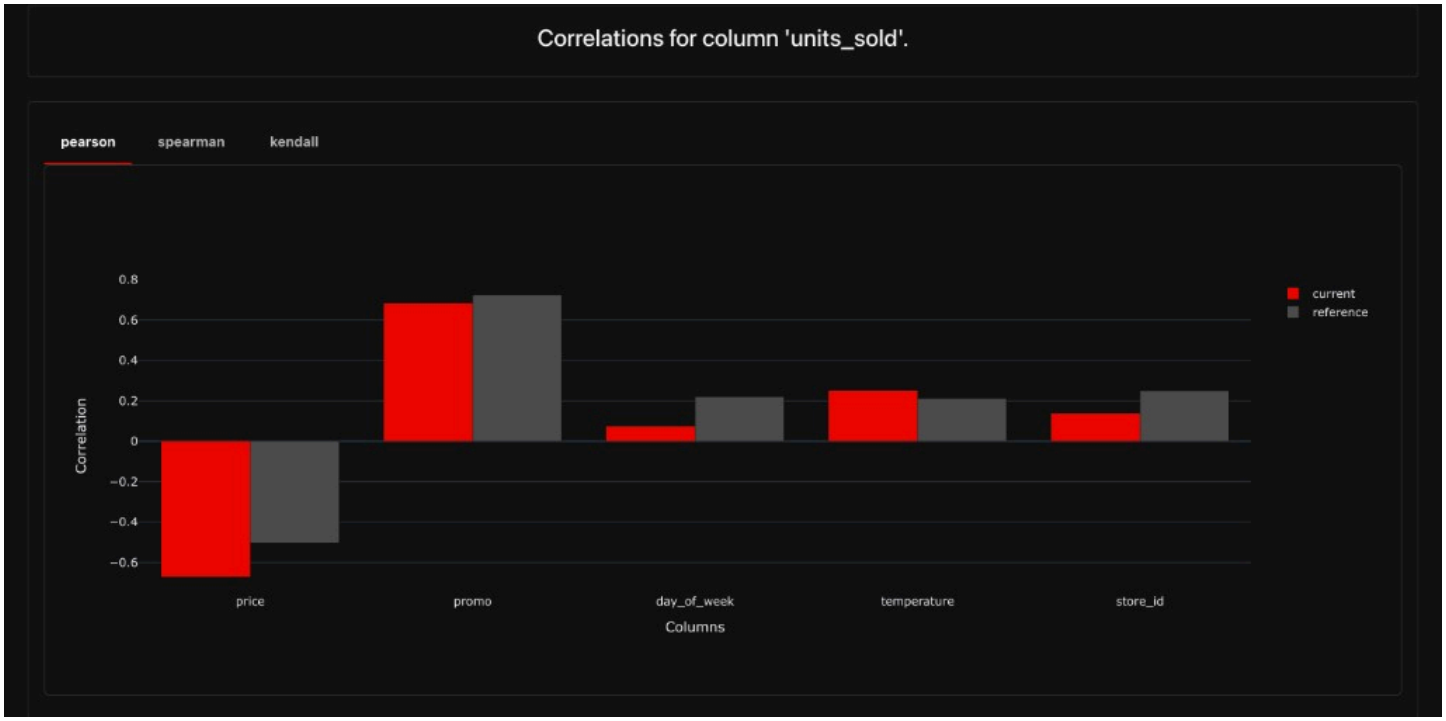
2. Report Screenshots - Drift in column units_sold (DATA DRIFT)



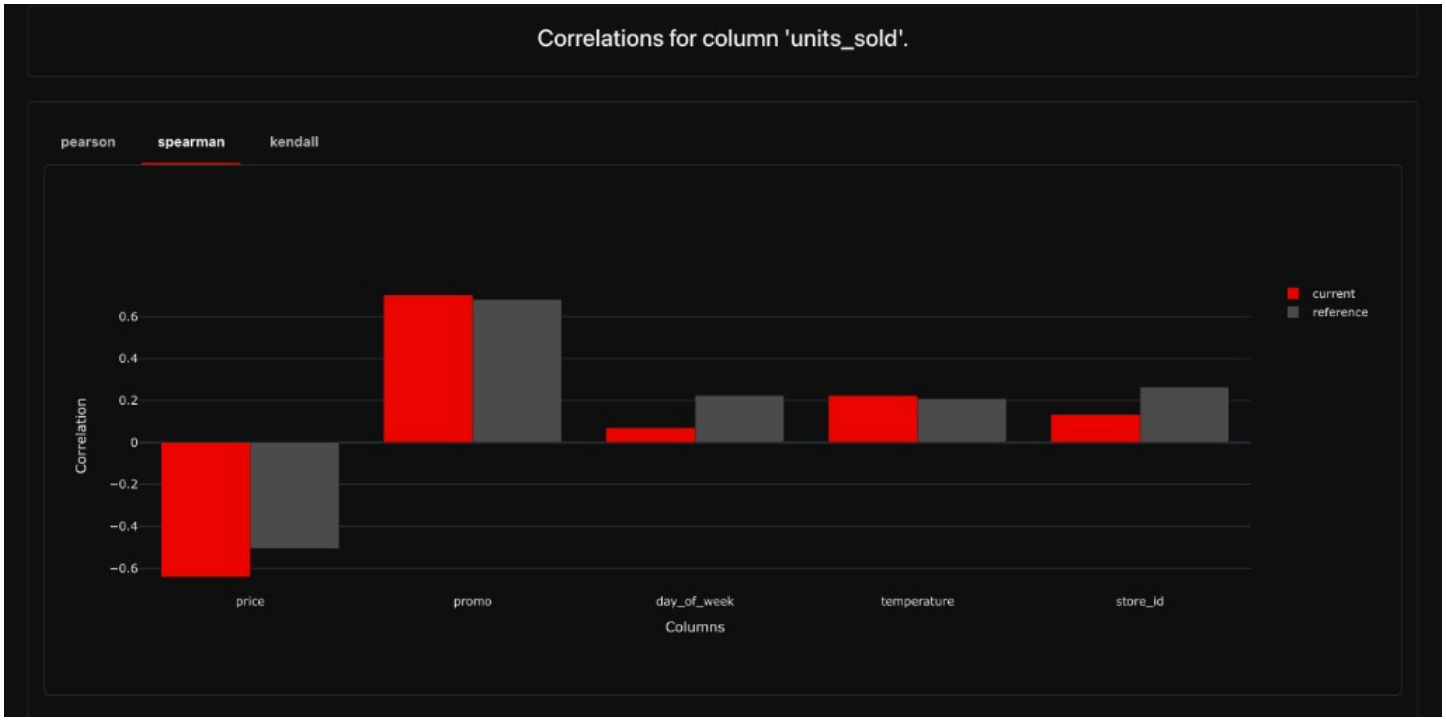
2. Report Screenshots - Column 'units_sold' Values



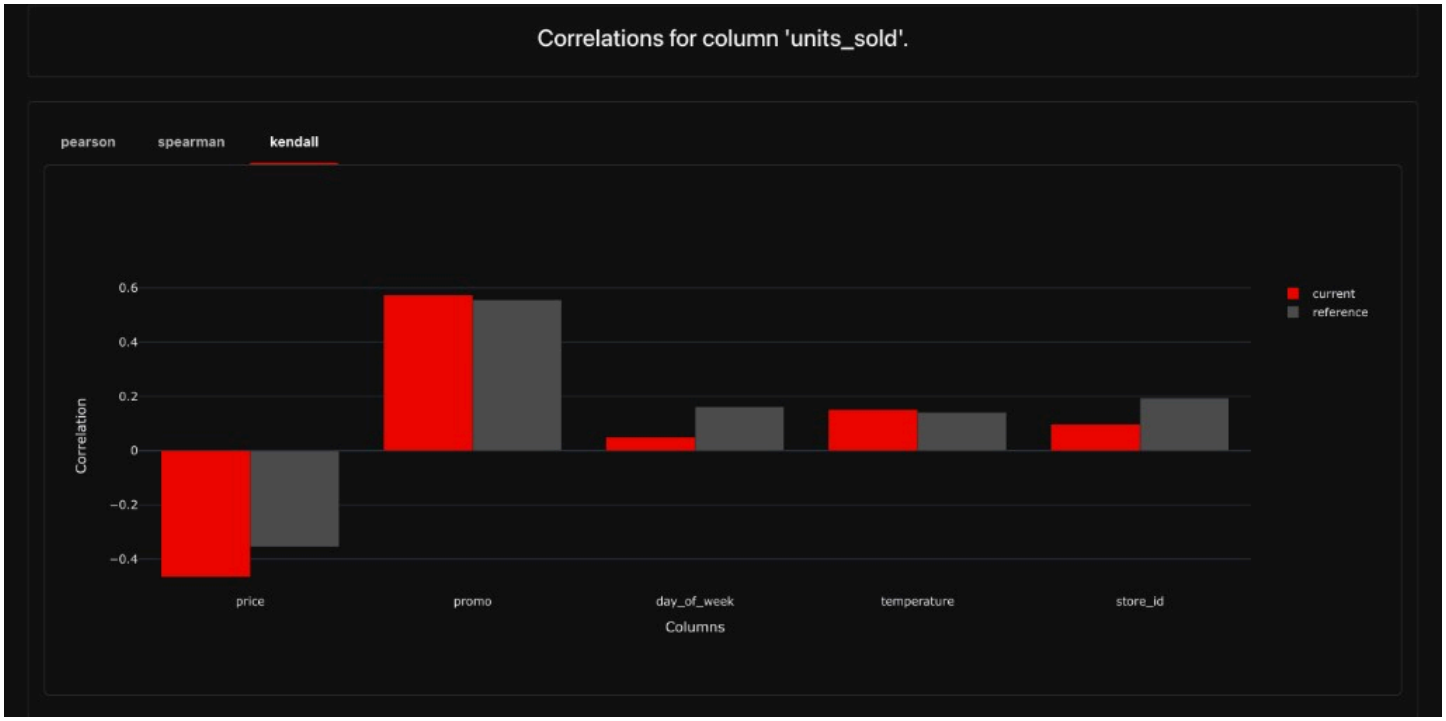
2. Report Screenshots - Correlations (Pearson) for units_sold



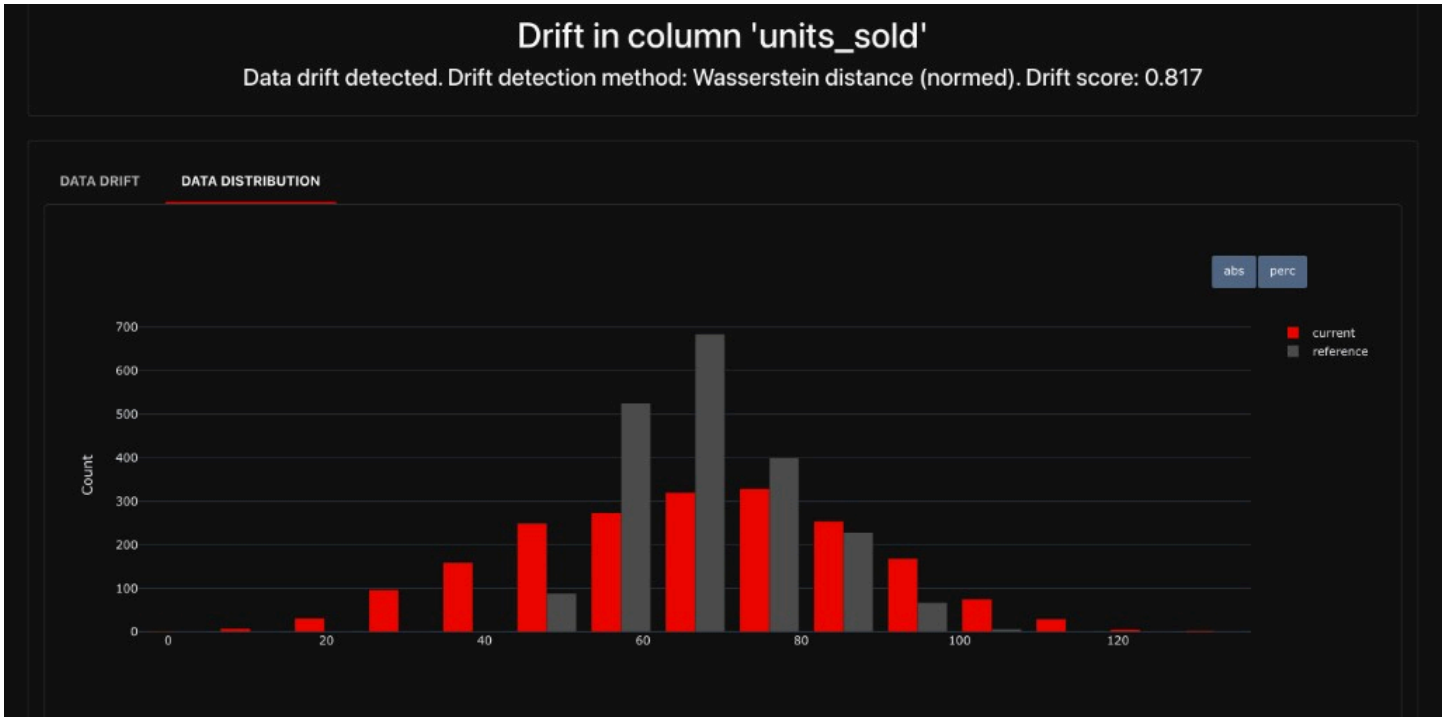
2. Report Screenshots - Correlations (Spearman) for units_sold



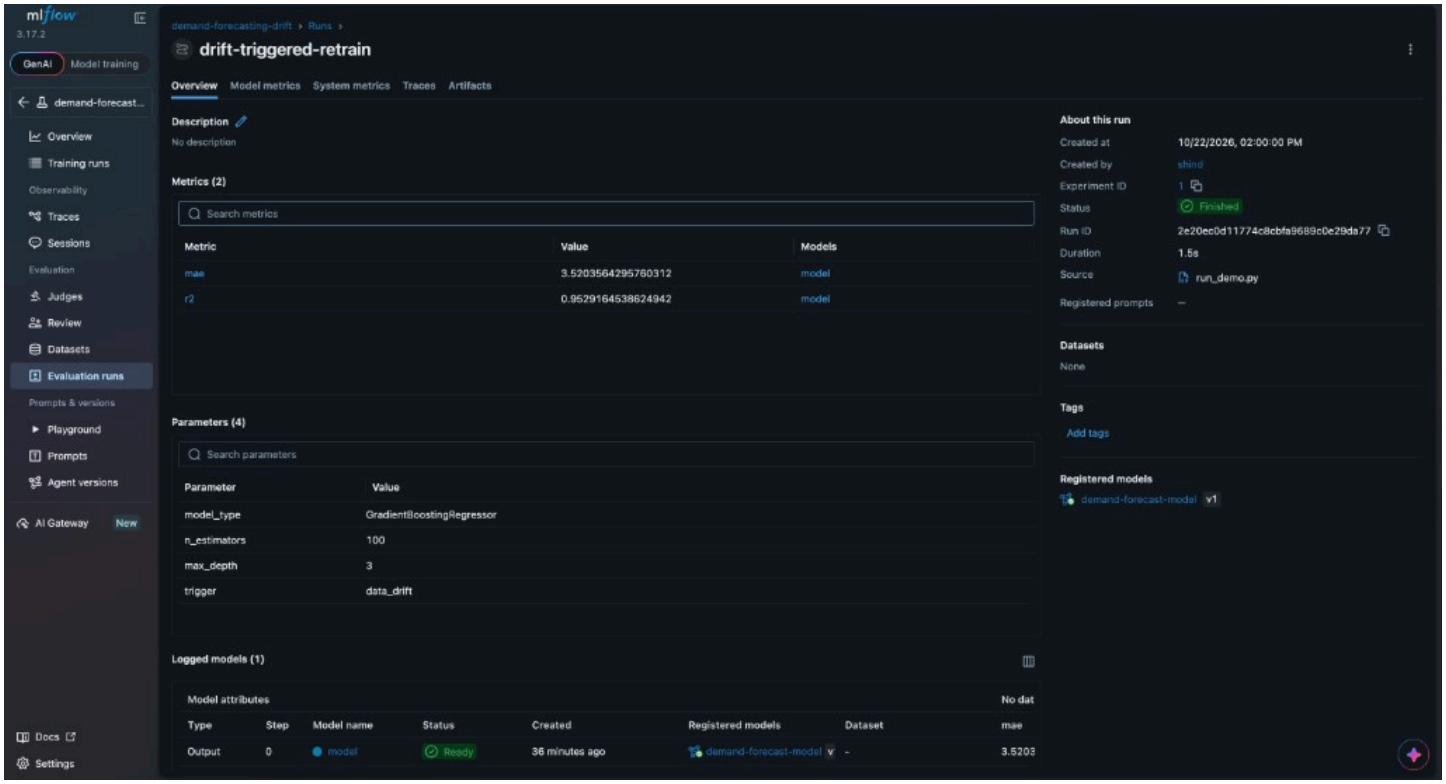
2. Report Screenshots - Correlations (Kendall) for units_sold



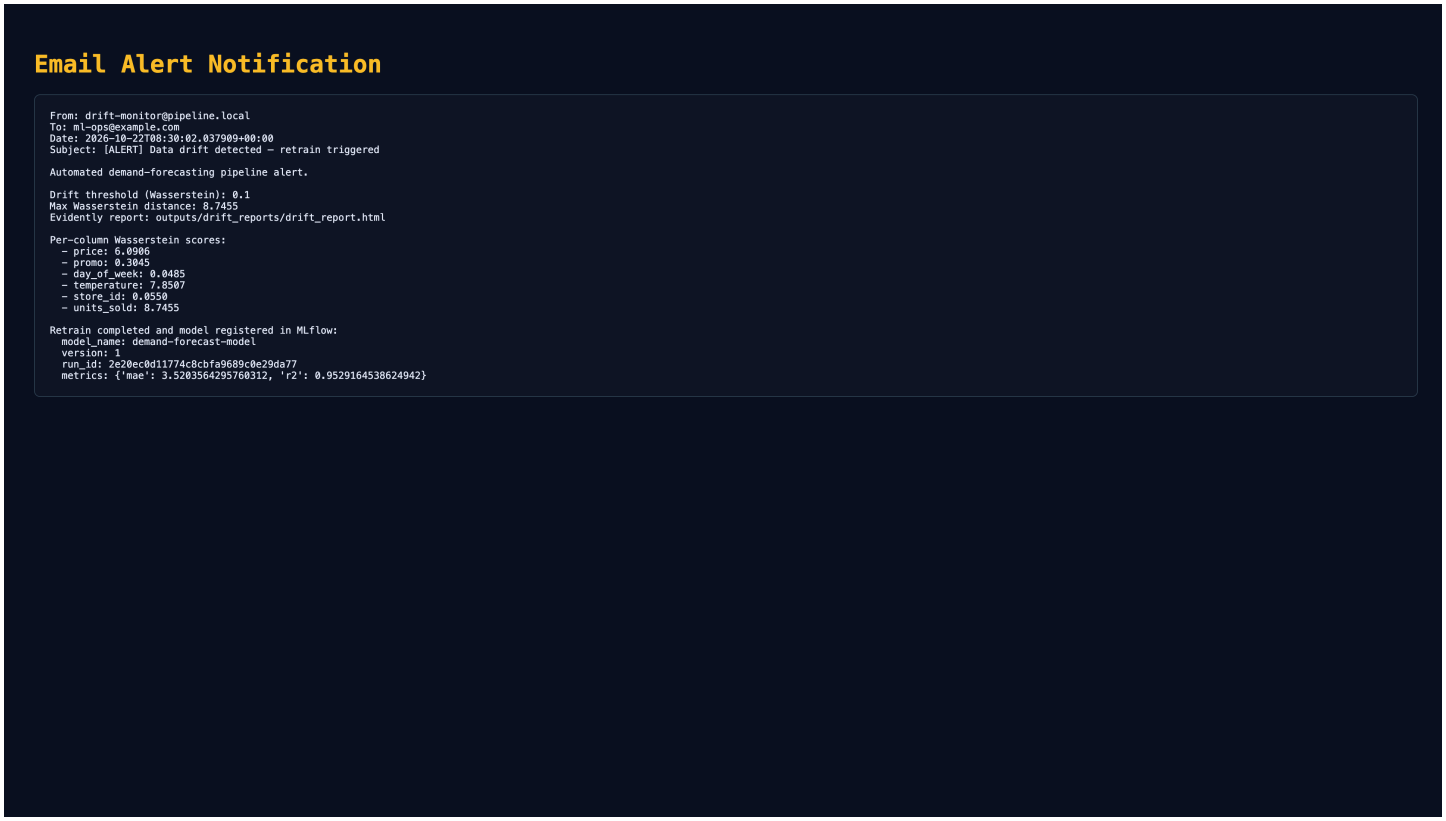
2. Report Screenshots - units_sold DATA DISTRIBUTION



3. MLflow Model Registry - drift-triggered-retrain run + demand-forecast-model v1



4. Email Alert Notification - retrain triggered



5. GitHub Repository - ADT23SOCB1617-drift-retraining-pipeline

